

curve values on a graph. The plotted graph uses different colours to highlight different state functioning, like linear, active and saturation mode for a transistor's characteristics, each of which is described by the graph legend. You can turn off the legend to make modifications to the graph.

For any circuit consisting of linear components, its equivalent circuit, as seen from any two points, can be obtained using Solve Elec. From the equivalent resistance of a set of resistors to Thevenin equivalent circuit, data is provided along with the formulae involved. For AC voltages and currents, oscilloscope feature can be exploited to view the signals as if on a real instrument.

What happens when frequency is changed. No electrical analysis is complete without analysing the circuit's frequency response. Solve Elec lets you trace this response using transfer functions and gains obtained for circuits working on an AC source. The necessary condition is that the circuit must contain a single input voltage source and only one voltmeter defining the output voltage.

A transfer function tool, when initiated, checks the circuit, analyses it and displays the results on the transfer function pane. For a resistor-capacitor (RC) filter circuit, the pane contains details like definitions of the function, values of the various parameters involved and finally ending with the cut-off frequency of the designed filter. Corresponding frequency response can be linear or logarithmic, and phase relations can also be displayed.

Set up with care, measure with ease

Solve Elec comes with a well-defined set of measuring instruments like the ammeter and the voltmeter, which can be included at necessary regions in the design. Computed

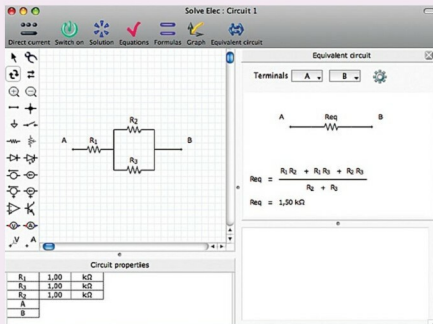


Fig. 2: Equivalent of the circuit given in Fig. 1

quantities are accompanied by their values and formulae used to calculate these, in a window called Solution. Calculated using a formal calculus engine, this comes in handy while manually verifying equations, drawing graphs and so on.

As always, potential reference is taken with respect to the ground terminal, and nodes can be named using point labels. Also available is a probe that can be used to measure the potential at a point, be it on a wire or at a terminal.

How about automating to reduce effort. Of course, the idea behind simulating a circuit is not only to verify its correctness, but also to compute parameters that cannot be measured directly. These could be something as simple as power dissipated by a resistor, or intense like the loading effect at the output.

Manually calculating these for each iteration of the design is even more tedious. By defining a formula for each of the necessary calculations and assigning it newly-defined quantities in the circuit, computations are automatically performed each time the circuit is solved.

If complex and time consuming,

In a nutshell

Latest version: Solve Elec 2.5

Developers: Physicsbox

Licence: Free

Purpose: To draw and analyse electrical circuits

Top features:

- Support for DC and AC
- Ease of use
- Will not simulate until perfect. On simulation, the program will inform you if the circuit does work

Languages supported: English and French

a wait window appears as a warning to you alongside an option to stop the process. The solution can check errors in signs, issues with homogeneity or incorrect application of standard laws like Kirchhoff's rules. It also spells out the states of circuit components, if different, along with explicit values of every parameter discussed.

Well, of course, the solution tab is a quick and sure-shot way of finding out if the circuit is solved. While the power-off button can turn the circuit off, any action like asking for a solution, equation or graph turns it on again.